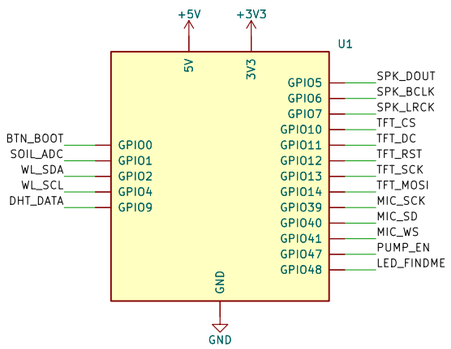
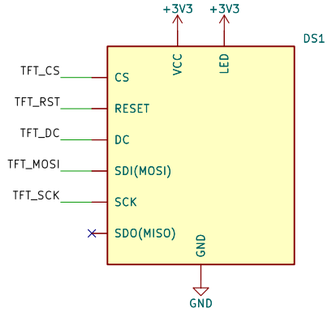
MCU



ESP32-S3-DevKitC-1 N16R8

ESP32-S3-DevKitC-1 (N16R8: 16MB flash, 8MB octal PSRAM). Powered via its 5V pin from the boost converter (see POWER); the onboard LDO supplies +3V3 to every module. Devkit USB-C = flash/console only (native USB-Serial-JTAG on GPIO19/20).
Reserved, never use: GPIO3/45/46 (strapping), 19/20 (USB), 22-25 (absent), 26-32 (flash), 33-37 (PSRAM), 43/44 (UART0).

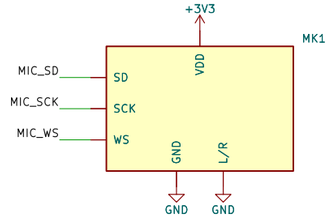
DISPLAY



ILI9341 240x320 SPI TFT

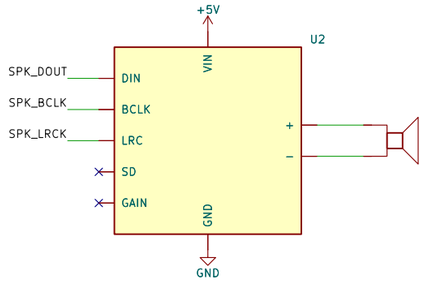
Write-only 4-wire SPI on SPI2_HOST (10MHz over jumper wires). Backlight LED tied to +3V3; move it to a free GPIO (e.g. GPIO15) if PWM dimming is wanted. Touch pins (T_*) unused.

AUDIO



INMP441 I2S MEMS mic

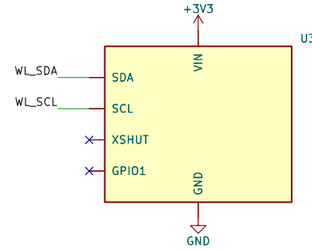
L/R to GND = left channel. 16kHz mono, 24-bit in 32-bit frames.



MAX98357A I2S amp

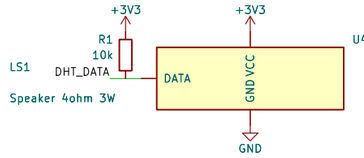
SD floating = (L+R)/2 output (breakout pull-up). GAIN floating = 9dB. If speaker hiss persists, check SD is not floating at threshold.

SENSORS



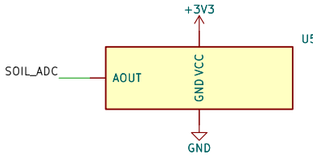
VL53L1X ToF (water level)

I2C_NUM_1, addr 0x29, own bus (GPIO2/4). Breakout has onboard 4.7k-10k pull-ups. Mounted at tank top facing down: EMPTY=46mm FULL=9mm are placeholders, calibrate.



DHT11 temp/humidity

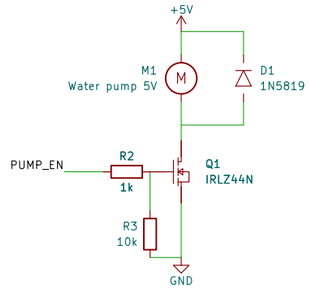
Single-wire bit-banged on GPIO9. Omit R1 if the 3-pin module already has an onboard pull-up.



Capacitive soil moisture v1.2

AOUT to GPIO1 = ADC1_CH0 (12dB attenuation). Needs dry/wet calibration.

ACTUATORS / INPUT



Water pump: IRLZ44N logic-level low-side switch, GPIO47 active-high. R2 1k limits gate charge current from the GPIO, R3 10k keeps the gate low during boot/reset. D1 clamps motor back-EMF.

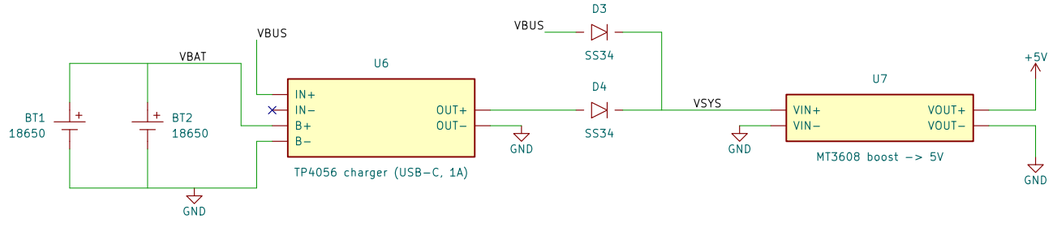


Find-me LED, GPIO48 active-high (led_indicator.cc).



External front button in parallel with the devkit BOOT button (GPIO0, strapping, active-low): click = toggle voice session, hold 10s = factory reset. Do not hold during power-on (enters download mode).

POWER



2x 18650 in PARALLEL (single-cell TP4056 rating). TP4056 module: USB-C on the module, IN+ pad = VBUS (tap it), IN- NC.
DIODE OR-ING (charge and load paths separated): D3 VBUS->VSYS, D4 cell->VSYS. USB present -> VSYS = 5V-Vf ~4.7V, D4 is reverse-biased (cell max 4.2V-Vf = 3.9V), so the load runs from USB and the cell sees only the TP4056 charge loop (correct CC/CV termination, boots with a dead cell). USB absent -> D4 conducts, VSYS = VBAT-Vf. Both must be Schottky (SS34: Vf ~0.3V at 1A, 3A rated) so the battery path loses as little as possible; never use 1N4007 here.
Cost of the diode path: ~0.3V and ~0.3W at 1A lost on battery power, and the boost sees a 0.3V lower input (brownout headroom shrinks). If that bites, swap D4 for a P-MOSFET ideal diode (AO3401, gate to VBUS, 100k to GND).
USB budget: charger (1A, Rprog=1.2k) + load share one 5V input -> use a 2A+ adapter or raise Rprog (3k = 400mA). Module USB-C needs CC pull-downs (5.1k) or a C-to-C cable gives no VBUS. Set MT3608 to 5.0V BEFORE connecting the devkit.
Planned UVLO (not built): insert between D4 cathode and VSYS so a drained cell cuts off instead of brownout-looping.

Module-level wiring (breakout boards on headers), not a bare-chip PCB design
Pin map source of truth: firmware-synora/main/boards/synora/config.h

Sheet: /
File: synora.kicad_sch

Title: Synora – plant-care voice companion

Size: A3	Date: 2026-11-04	**Rev: 1**
KiCad E.D.A. 10.0.6		Id: 1/1